Review: The Greeks

For small changes in the share price, the volatility, the calendar time (years), and the rate, the premium $\mathcal{P}_\star$ of a call ($\star = c$) or put ($\star = p$) changes by:

$$\begin{aligned}\Delta\mathcal{P}_\star &\approx \Delta_\star\,\Delta S + \tfrac{1}{2}\,\Gamma_\star\,(\Delta S)^2 + \nu_\star\,\Delta\sigma + \Theta_\star\,\Delta t + \rho_\star\,\Delta g_y \\ &= \underbrace{(\Delta_\star + \tfrac{1}{2}\,\Gamma_\star\,\Delta S)\Delta S}_{\text{share price}} + \underbrace{\nu_\star\,\Delta\sigma}_{\text{volatility}} + \underbrace{\Theta_\star\,\Delta t}_{\text{time}} + \underbrace{\rho_\star\,\Delta g_y}_{\text{rate}}\end{aligned}$$

- Delta and Gamma: the slope and curvature of the premium in S.
- Vega, Theta, and Rho: sensitivities to the volatility σ, the calendar time t, and the risk-free rate g_y, quoted as $\Theta_\star/365$ per day and $\nu_\star/100$, $\rho_\star/100$ per point.
- Hedge ratio: one contract covers 100 shares, so it is equivalent to $100 \cdot \Delta_\star$ shares.

Example 1: CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

Review: POE and POP

The POE is the probability of finishing in the money by more than 0.01 USD/share; the POP is the probability of a positive profit at expiration, beyond the breakeven $S^{\ddagger} = K \pm \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_\star$ (+ call, − put):

$$\text{POE} = \begin{cases} \mathbb{P}\left(S(T) > K + 0.01\right) & \text{call} \\ \mathbb{P}\left(S(T) < K - 0.01\right) & \text{put} \end{cases}$$

$$\text{POP} = \begin{cases} \mathbb{P}\left(S(T) > S^{\ddagger}\right) & \text{bought a call or sold a put} \\ \mathbb{P}\left(S(T) < S^{\ddagger}\right) & \text{sold a call or bought a put} \end{cases}$$

where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$; we simulate $S(T)$ under our belief $\mathbb{P}$. However, the POE $\sim |\Delta_\star|$, the Delta quoted by trading platforms under $\mathbb{Q}$.

Review: Computing POE and POP by Simulation

Initialize: given $S(0)$, K, the premium $\mathcal{P}_\star$ of the call or put, T (years), the drift μ and volatility σ under $\mathbb{P}$, g_y, and the number of samples M. Compute the breakeven $S^\ddagger$.

1. For each sample $j = 1, 2, \ldots, M$ **do**:
   a. Draw $Z_j \sim \mathcal{N}(0,1)$.
   b. Compute $S_j(T) \leftarrow S(0) \exp\left[(\mu - \sigma^2/2)\,T + \sigma\sqrt{T}\,Z_j\right]$.
2. Compute the fraction of samples at or below each share price x:
   $\hat{F}(x) \leftarrow \frac{1}{M} \cdot \#\left\{j : S_j(T) \le x\right\}$.

Review: Pseudocode for POE and POP

3. If the contract is a call, **then** POE $\leftarrow 1 - \hat{F}(K + 0.01)$. Otherwise, POE $\leftarrow \hat{F}(K - 0.01)$.
4. Compute the probability of profit:
   - If we bought a call or sold a put, **then** POP $\leftarrow 1 - \hat{F}(S^{\ddagger})$.
   - If we sold a call or bought a put, **then** POP $\leftarrow \hat{F}(S^{\ddagger})$.

With $\mu = g_y$ and σ set to the implied volatility, the same steps give the probabilities under $\mathbb{Q}$. These are the probabilities inside the market price, not our forecast.

Example 2: CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Company Profile: Akuna Capital

Akuna Capital: an options market maker, headquartered in Chicago since 2011.

- Options market making: it posts prices to buy and sell, trading with its own money (a proprietary trading firm).
- Its own models: it designs its own trading systems and models.
- Teaching options: a free Options 101 course on payoff diagrams and the Greeks.

Connection: a market maker may take the other side of one or both legs of the spread we sell. The quoted Delta comes from its premiums, so $|\Delta_\star|$ approximates each leg's POE under $\mathbb{Q}$, not under our forecast $\mathbb{P}$.